DIODE CIRCUITS

SEMICONDUCTOR DIODES

prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

50%

WEEKLY

50%

EXAM

100%

LABORATORY

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

DC Input

- OR Gate and AND Gate

Sinusoidal Input

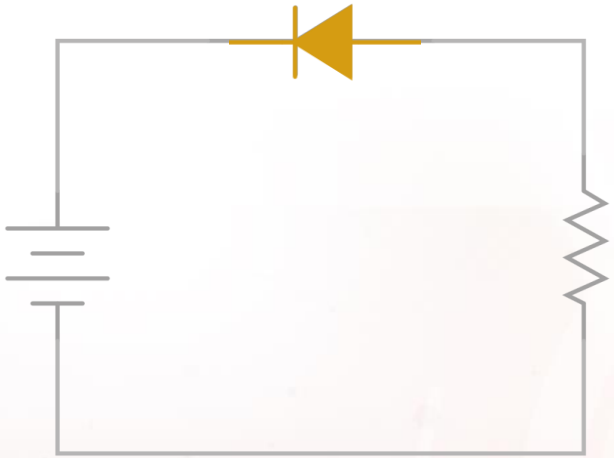
- Half-Wave Rectifier
- Full-Wave Center-Tapped Rectifier
- Full-Wave Bridge-Type Rectifier

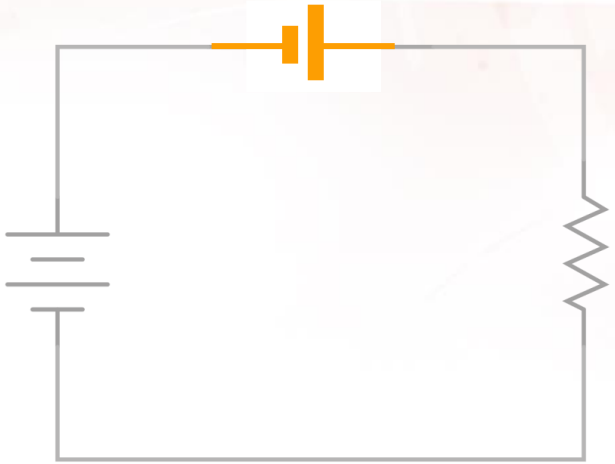
DC INPUT

DIODE CIRCUITS

ON DIODE

"OFF" DIODE



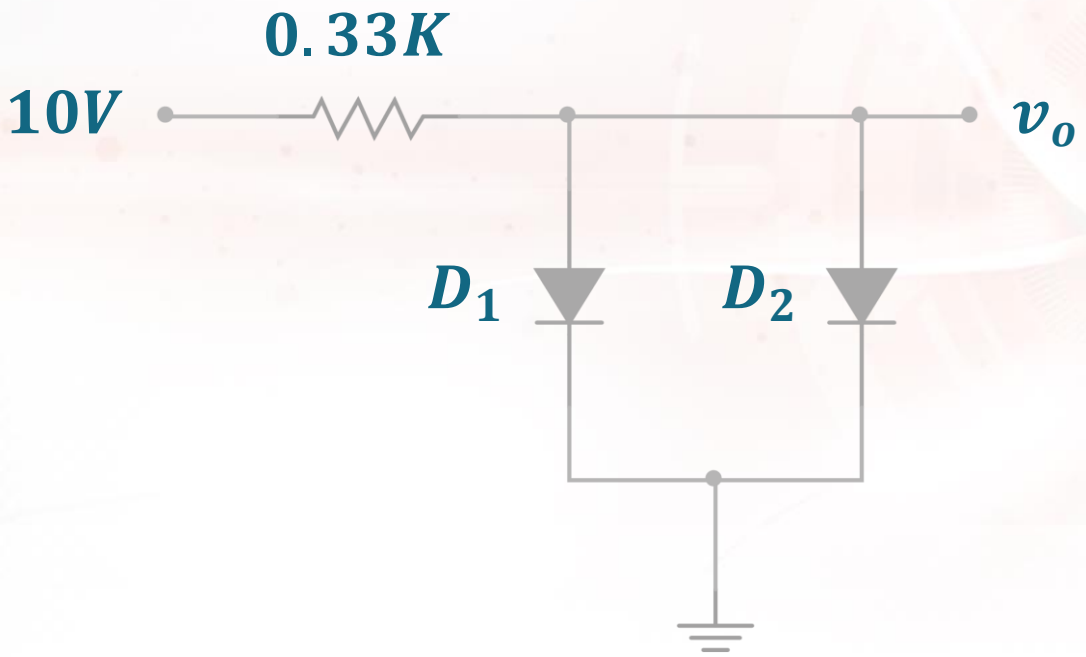


Determine the v_D, v_R, i_D for the given **network**.

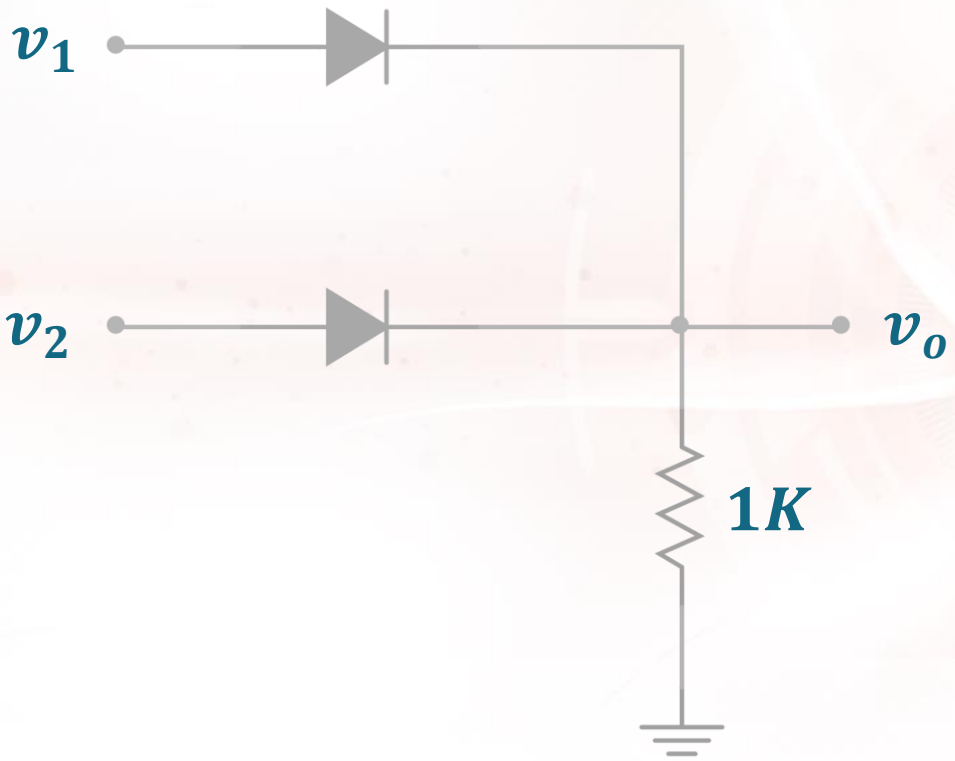
Determine the v_D, v_R, i_D for the given **network**.

Determine the v_o and i_D for the given **network**.

Determine the v_o, i_R, i_{D1}, and i_{D2} for the given **network**.

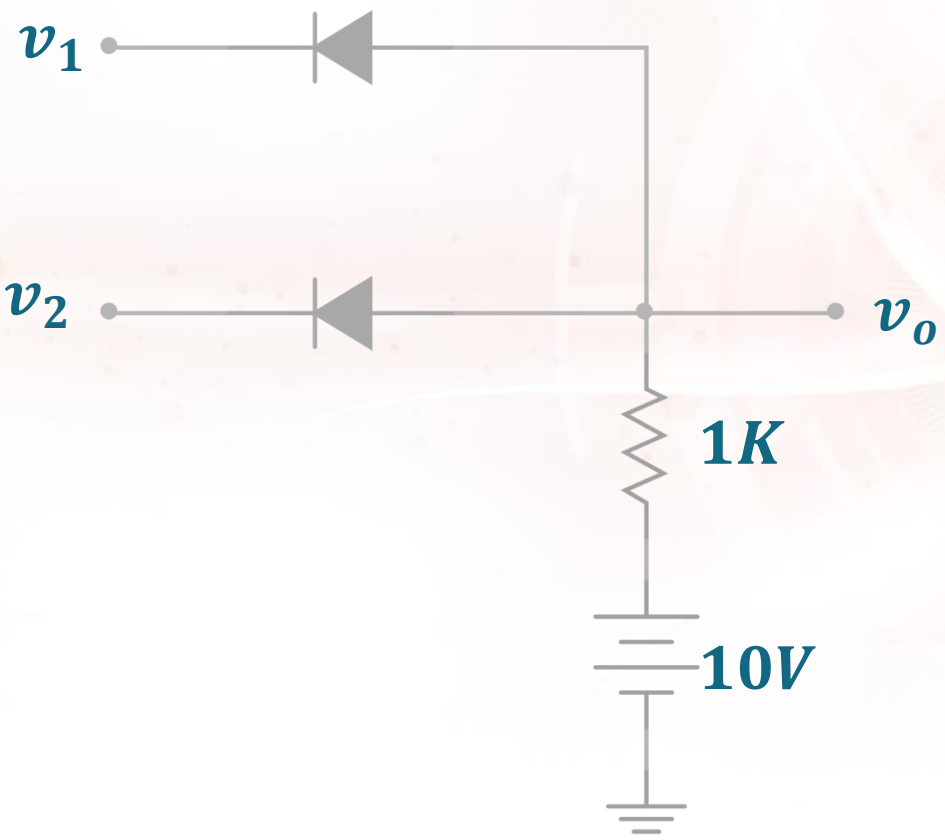


POSITIVE LOGIC OR GATE



v_1	v_2	v_o
0V	0V	
0V	10V	
10V	0V	
10V	10V	

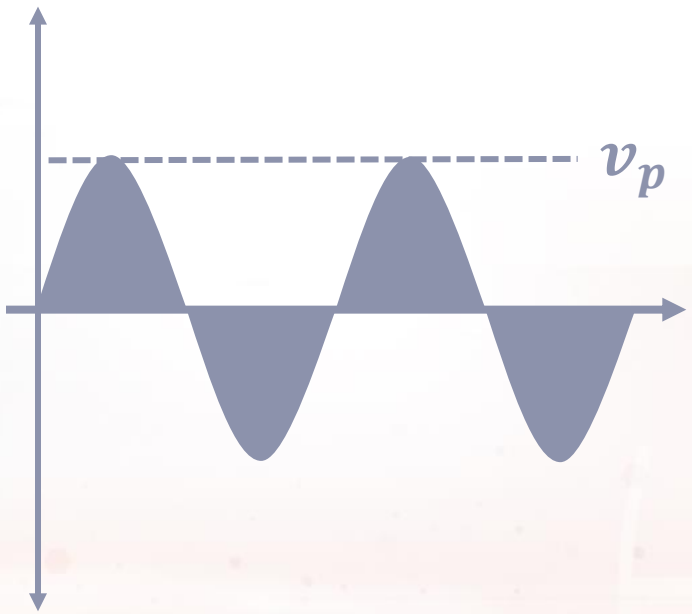
POSITIVE LOGIC AND GATE

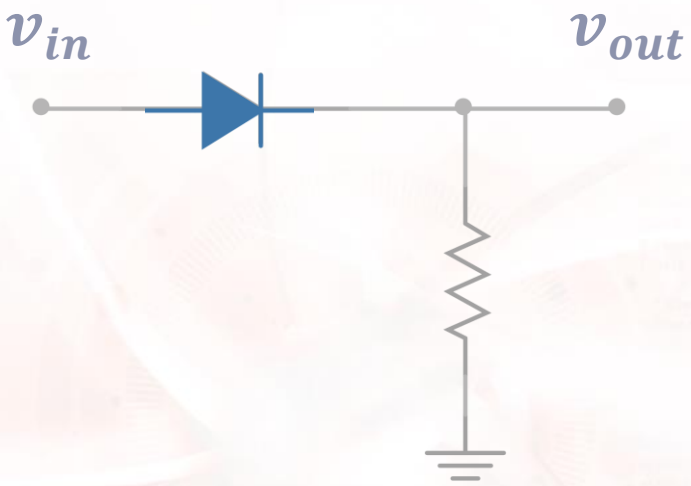


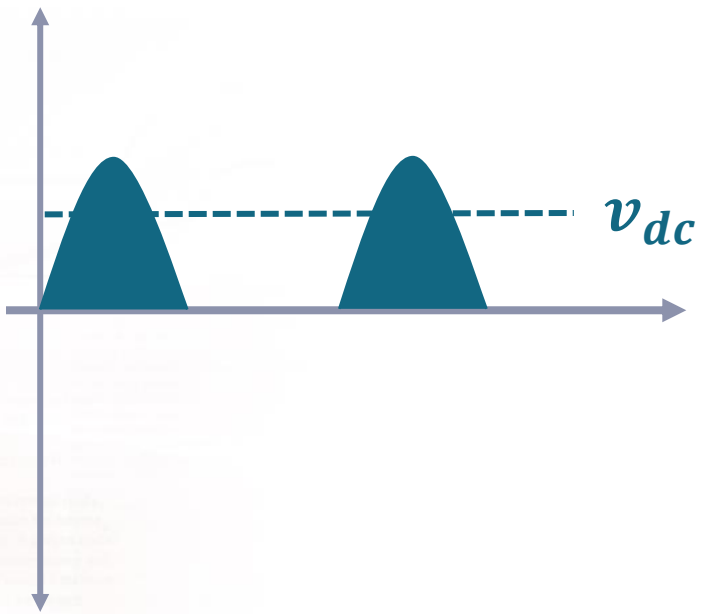
v_1	v_2	v_o
0V	0V	
0V	10V	
10V	0V	
10V	10V	

SINUSOIDAL INPUT DIODE CIRCUITS

HALF-WAVE RECTIFICATION







Output Frequency,

$f_{out} = f_{in}$

Average Voltage,

$v_{dc} = 0.318(v_p - v_T)$

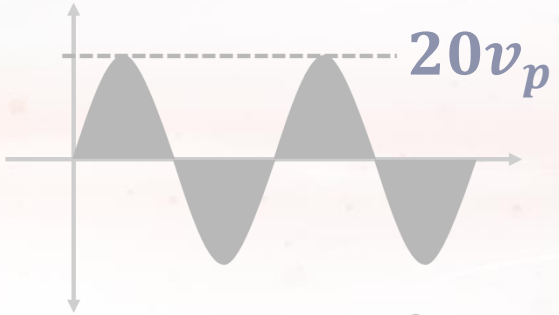
Threshold Voltage,

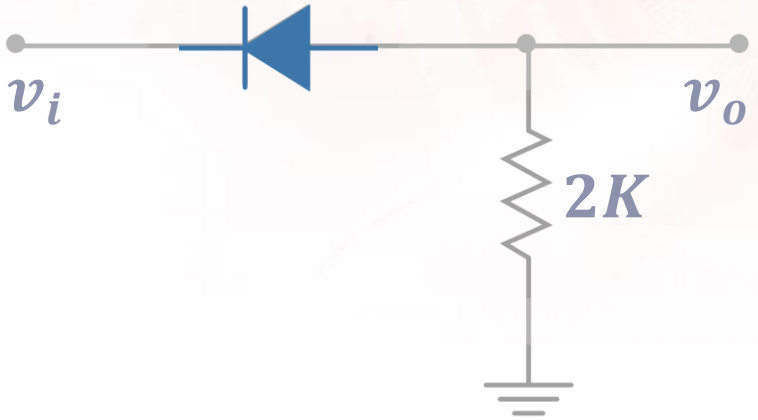
$v_T = 0 \quad (ideal)$

$v_T = 0.7 \quad (Si)$

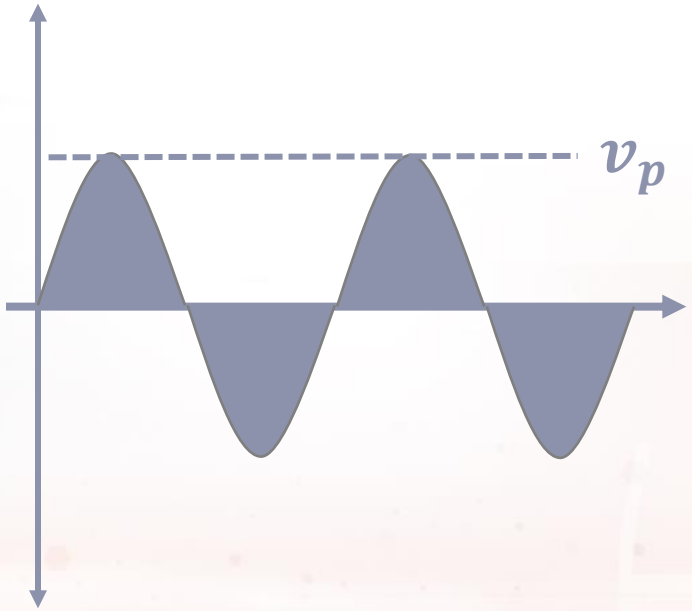
$v_T = 0.3 \quad (Ge)$

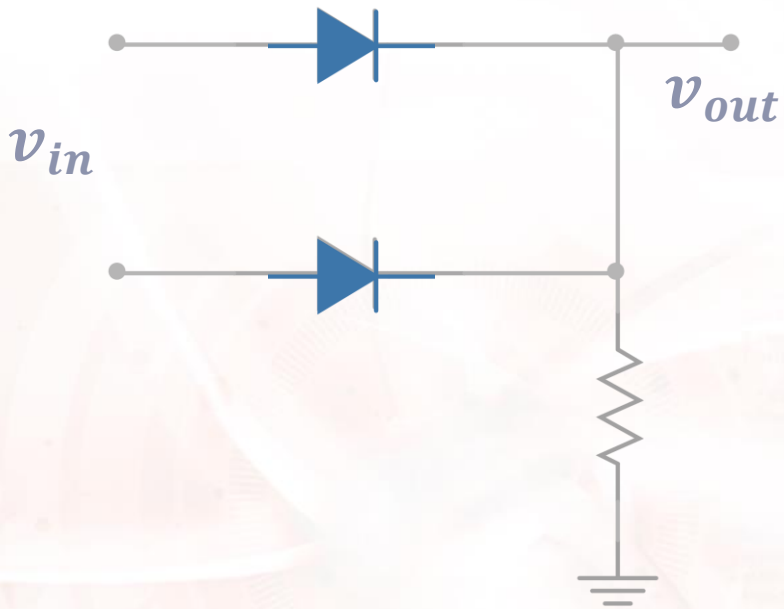
Sketch the output v_o and determine the **dc level** of the output for the given network.

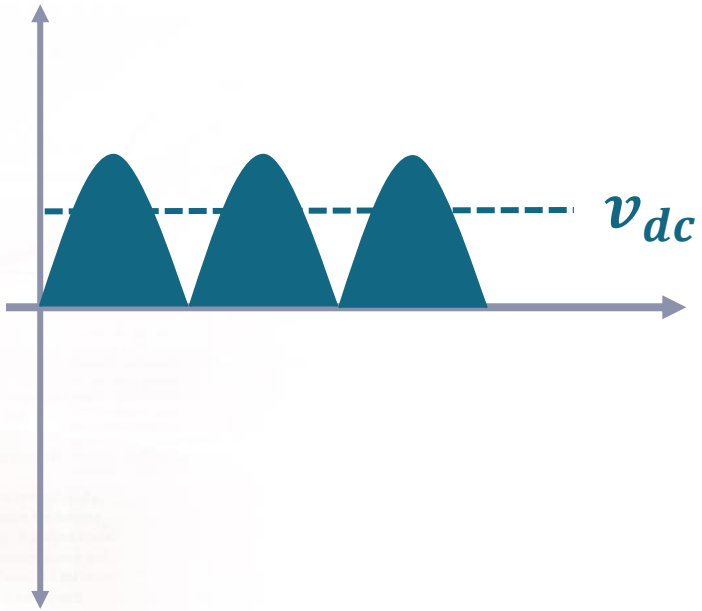




CENTER-TAPPED







Output Frequency,

$$f_{out} = 2f_{in}$$

Average Voltage,

$$v_{dc} = 0.636(v_p - v_t)$$

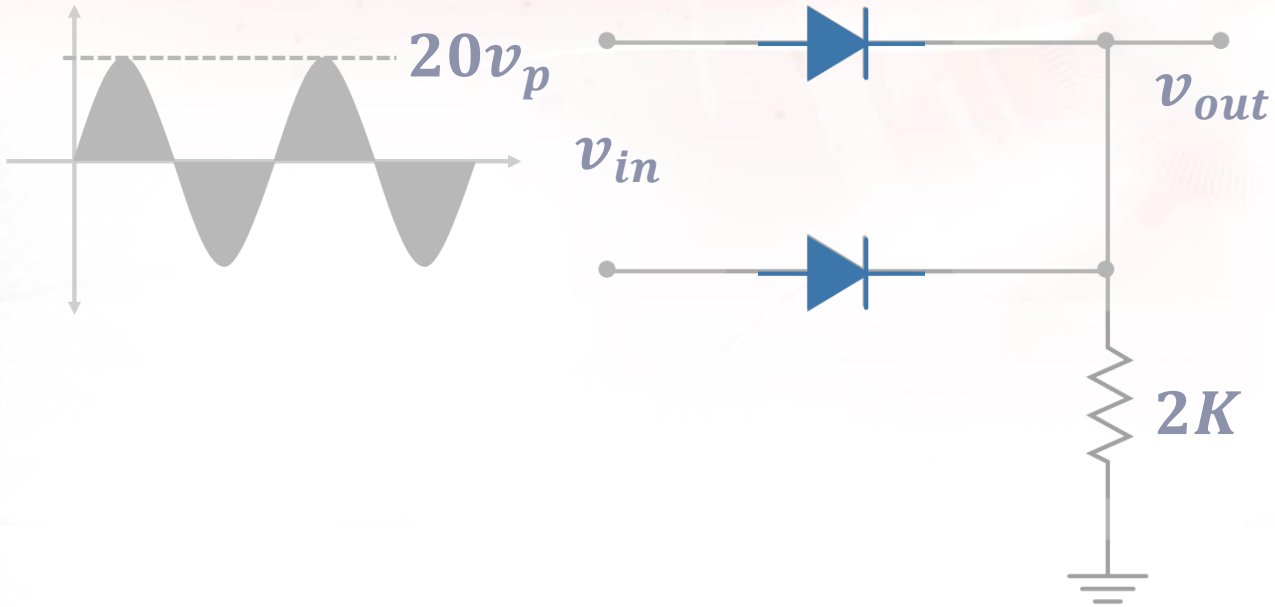
Threshold Voltage,

$v_t = 0 \quad (ideal)$

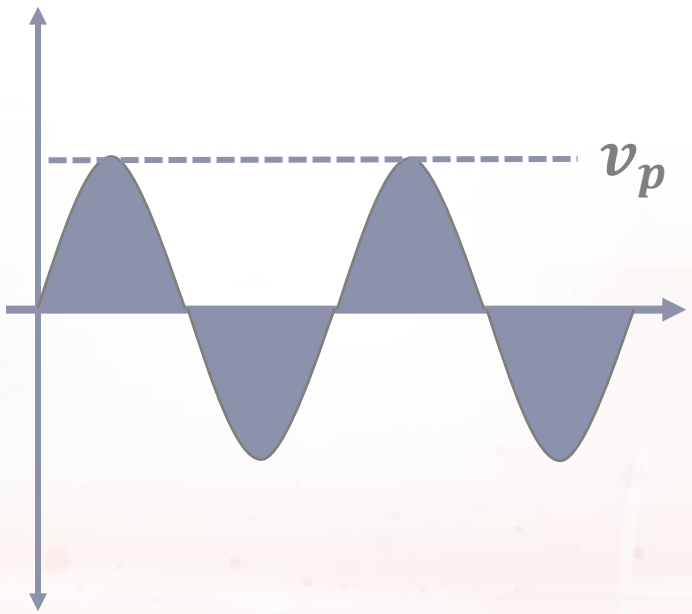
$v_t = 0.7 \quad (Si)$

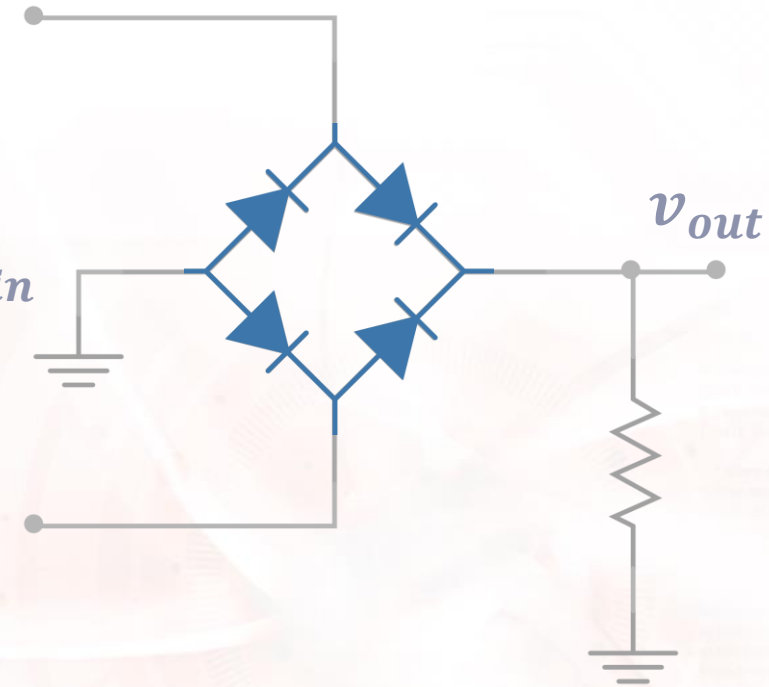
$v_t = 0.3 \quad (Ge)$

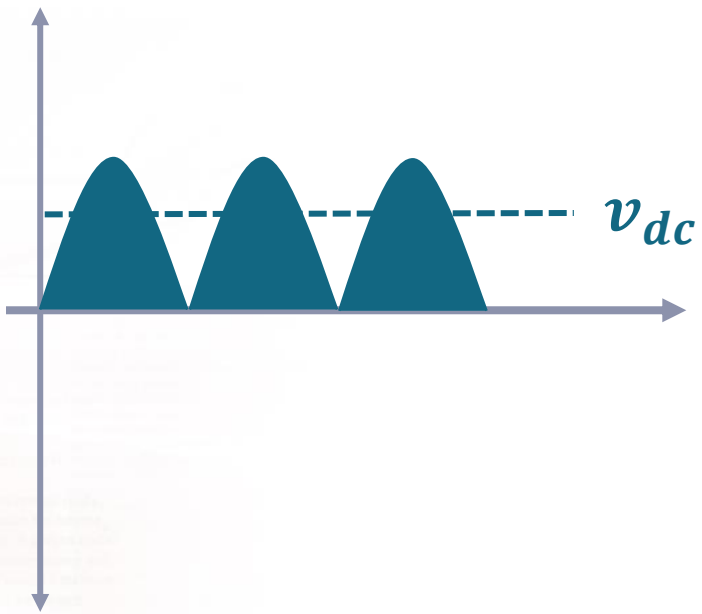
Sketch the output v_o and determine the **dc level** of the output for the given network.



BRIDGE TYPE







Output Frequency,

$$f_{out} = 2f_{in}$$

Average Voltage,

$$v_{dc} = 0.636(v_p - 2v_t)$$

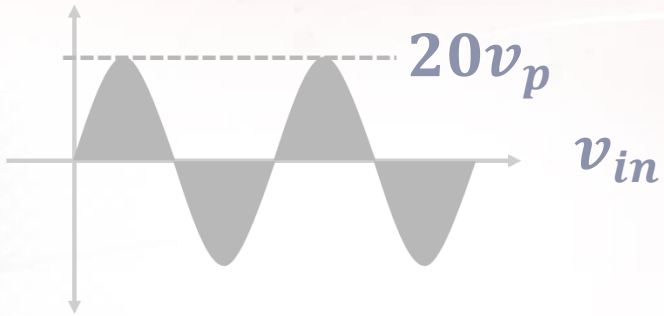
Threshold Voltage,

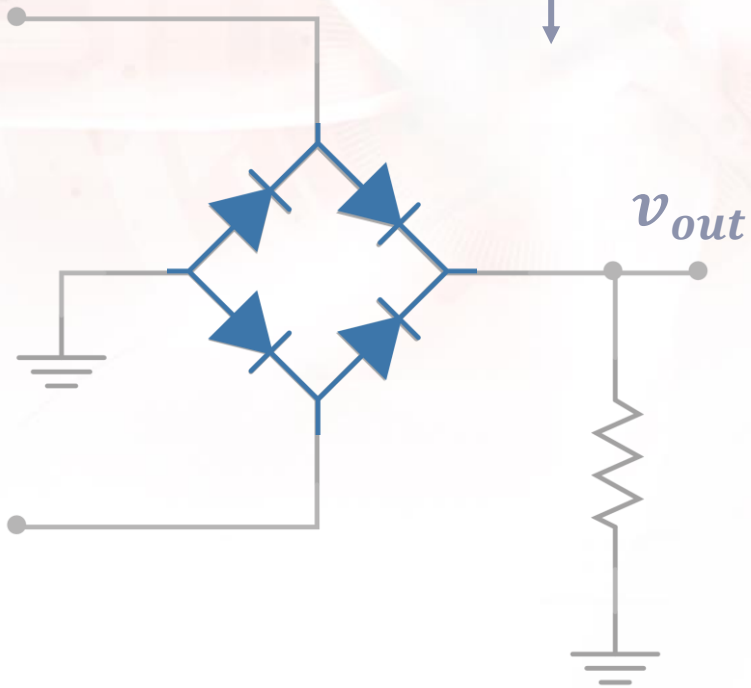
$v_t = 0 \quad (ideal)$

$v_t = 0.7 \quad (Si)$

$v_t = 0.3 \quad (Ge)$

Sketch the output v_o and determine the **dc level** of the output for the given network.





LABORATORY